JECDE
Journal of Electrical, Computer and Digital Engineering

TEMPLATE DEMONSTRATION AUTHOR MANUSCRIPT

Designing reproducible engineering research: a guide to the JECDE article format

JECDE Editorial Design

Author template development edition; no editorial appointment is implied.

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ABSTRACT

This document demonstrates the proposed article design for the Journal of Electrical, Computer and Digital Engineering. It illustrates title hierarchy, a full-width abstract, two-column body text, numbered equations, vector figures, formal tables and an algorithm with a coloured header. The content is instructional rather than a published research article. Mathematical examples and graphics explain formatting conventions; they do not represent experimental findings. Publication identifiers and acceptance dates are deliberately absent because none have been assigned.

Keywords: electrical engineering; electronics; computing; communications; artificial intelligence; cybersecurity; digital technologies

1 Purpose and scope

This template provides a consistent visual language across the journal's seven subject tracks. Electrical engineering, electronics, computing, communications, artificial intelligence, cybersecurity and digital technologies share the same article structure while retaining discipline-specific methods and notation.

Design demonstration

This is an author template, not a published article. No ISSN, DOI, volume, acceptance date or indexing status is claimed. Replace all instructional content before submission.

1.1 Structure and contribution

An original research article should state a precise problem, explain its novelty in context, define a reproducible method and support its conclusions with evidence. A review article should describe its literature-selection method and provide critical synthesis.

Use numbered sections and concise subsection titles. Define acronyms at first appearance. Cite figures, tables and equations in the text before discussing their implications.

2 Mathematics and notation

Mathematical notation should be consistent throughout the manuscript. For example, a generic engineering allocation problem can be expressed as

$$\underset{\boldsymbol{x}}{\text{minimize}} \quad J(\boldsymbol{x}) = \sum_{i=1}^{N} w_i f_i(x_i), \tag{1}$$

$$\text{subject to} \quad \sum_{i=1}^{N} x_i \le B, \quad x_i \ge 0. \tag{2}$$

Here, x_i denotes the resource allocated to component i, B is the total budget, w_i is a dimensionless importance weight and f_i is a problem-specific cost function. Equations (1) and (2) are formatting examples; a research manuscript must justify its own assumptions and units.

2.1 Reproducibility

Report software, data, parameters and evaluation conditions. Distinguish analytical, simulated and measured results. Table 1 summarizes track-specific reporting checks.

3 Figures and visual evidence

Use vector graphics where practical and ensure labels remain readable at the final column width. Figure 1 demonstrates a compact workflow with consistent shapes and line weights.

3.1 Tables

Table 2 uses formal horizontal rules rather than shaded cells or a dense grid. Keep units in column headings and explain abbreviations in the surrounding text.

Table 1. Recommended checks across the seven subject tracks.

Track	Typical evidence	Reporting emphasis
Electrical engineering	Analysis, simulation or measurements	Units, operating conditions, stability and losses
Electronics	Circuit models and device measurements	Component values, fabrication and uncertainty
Computing	Benchmarks and software artifacts	Environment, complexity and reproducibility
Communications	Link or network evaluation	Channels, traffic, baselines and metrics
Artificial intelligence	Training and independent testing	Data splits, leakage prevention and robustness
Cybersecurity	Threat models and security evaluation	Adversary capabilities and ethical handling
Digital technologies	Integrated system evaluation	Architecture, deployment and interoperability

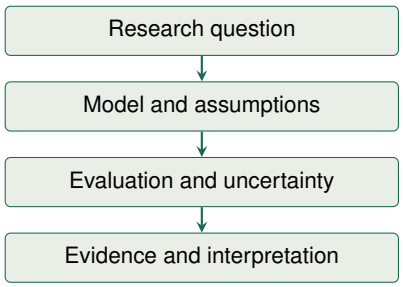


Figure 1. A research workflow illustrated with editable vector artwork.

Table 2. Article components and their purpose.

Component	Purpose
Abstract	Problem, approach and supported findings
Methods	Assumptions and reproducible procedure
Results	Evidence, comparisons and uncertainty
Discussion	Interpretation and limitations

4 Algorithms

Use an explicit input/output structure and meaningful operations. The numbered example below demonstrates the journal's header-bar style. It is a manuscript-checking procedure, not a novel research algorithm.

Algorithm 1. Manuscript readiness check

Input: manuscript M, checklist C.
Output: unresolved issues U.
1. Initialize $U \leftarrow \varnothing$.
2. For each requirement $c \in C$, inspect the relevant part of M.
3. If c is not satisfied, append an evidence-based note to U.
4. Verify figures, equations, references and declarations.
5. Return U for author revision.

5 Reporting and interpretation

A results section should explain how the evidence answers the research question. Report the evaluation conditions and appropriate baselines, then discuss why changes occur. Include uncertainty or repeatability information where applicable.

5.1 References

References should be accurate, relevant and cited in order of first appearance. A sample citation to a typesetting reference is provided here [1]. Replace it with sources appropriate to the submitted research.

6 Wide material and final checks

Two-column figures and tables are supported by the standard `figure*` and `table*` environments. Place wide content near its first discussion and allow the typesetting engine to find a suitable position.

7 Conclusion

The JECDE format combines a recognizable opening page with conventional scholarly typesetting. Authors should preserve readability, provide inspectable evidence and ensure that presentation supports the scientific contribution.

Declarations

Funding: No external funding supported this template demonstration.

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Data and code availability: This document contains no experimental dataset. Its editable LaTeX source accompanies the template package.

Author contributions: The document is a design and formatting demonstration, not a research authorship record.

References

[1] L. Lamport, *LaTeX: A Document Preparation System*, 2nd ed. Addison-Wesley, 1994.